

TASTY RECIPES

SOFTWARE REQUIREMENTS SPECIFICATION

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*Requirements structure follows professional software requirements engineering conventions and
ISO/IEC/IEEE 29148 principles.*

TABLE OF CONTENTS

Chapter 1: Introduction	1
1.1 Introduction	1
1.2 Project Background	1
1.3 Problem Definition	1
1.4 Project Objectives	1
1.5 Project Scope	1
1.6 User Needs	1
1.7 Constraints and Assumptions	2
Chapter 2: Introduction to Proposed System	3
2.1 Introduction	3
2.2 Proposed Solution	3
2.3 Target Users	3
2.4 Product Features	3
2.5 Operating Environment	4
2.6 System Context	4
Chapter 3: Requirement Specifications	5
3.1 Introduction	5
3.2 Functional Requirements	5
3.3 Fully Dressed Use Cases	5
3.4 External Interface Requirements	7
3.5 Non-Functional Requirements	7
3.6 Acceptance Criteria	8
Chapter 4: Design Specifications	9
4.1 Introduction	9
4.2 System Architecture	9
4.3 Website Structure	9
4.4 Use Case Diagram	10
4.5 Block Diagram	11
4.6 Class Diagram	11
4.7 Sequence Diagram	12
4.8 File and Folder Design	12
Chapter 5: Test Specifications	13
5.1 Introduction	13
5.2 Test Strategy	13
5.3 Test Cases	13
5.4 Requirements Traceability Matrix	14
Chapter 6: Deployment and Conclusion	15
6.1 Deployment Requirements	15
6.2 Deployment Procedure	15
6.3 Limitations	15
6.4 Future Enhancements	15
6.5 Conclusion	16

Appendix A: Glossary.....	17
Bibliography	18

LIST OF TABLES

Table 2.1: Product Features

Table 3.1: Functional Requirements

Table 3.2: UC-01 Browse Recipes

Table 3.3: UC-02 View Recipe Details

Table 3.4: UC-03 View About Information

Table 3.5: UC-04 Use Contact Form

Table 3.6: Non-Functional Requirements

Table 5.1: Test Cases

Table 5.2: Requirements Traceability Matrix

LIST OF ILLUSTRATIONS

Figure 2.1: System Context Diagram

Figure 4.1: Website Navigation / Site Map

Figure 4.2: Use Case Diagram

Figure 4.3: Activity Diagram

Figure 4.4: Conceptual Class Diagram

Figure 4.5: Sequence Diagram - View Recipe Details

LIST OF ABBREVIATIONS

Abbreviation	Term
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
UI	User Interface
UX	User Experience
HTML	HyperText Markup Language
HTTPS	Hypertext Transfer Protocol Secure
UC	Use Case
TC	Test Case
RTM	Requirements Traceability Matrix

INTRODUCTION

1.1 Introduction

Tasty Recipes is a beginner-level multi-page recipe website designed to demonstrate correct HTML5 structure, navigation, content organization, accessibility basics, documentation, testing, version control, and deployment. The project intentionally focuses on HTML before introducing CSS or JavaScript in later projects.

1.2 Project Background

A small recipe website is suitable for learning core web concepts because it combines headings, paragraphs, images, links, lists, forms, semantic sections, and multiple pages in one understandable domain. The project also provides a practical example that can be reviewed by students learning how requirements move from documentation into implementation.

1.3 Problem Definition

Beginning developers often build pages without first defining requirements, file structure, acceptance criteria, or testing. This can lead to inconsistent navigation, broken links, inaccessible markup, and unclear project scope. Tasty Recipes addresses this learning problem by using a documented, requirements-driven workflow from specification through deployment.

1.4 Project Objectives

- Develop a complete multi-page static website using semantic HTML5.
- Provide clear recipe browsing and recipe detail pages.
- Apply meaningful headings, lists, links, labels, and image alternative text.
- Maintain an organized repository and project directory.
- Verify requirements through documented test cases.
- Deploy the completed website on a public static hosting platform.

1.5 Project Scope

Version 1.0 includes Home, Recipes, individual Recipe Details, About, and Contact pages. It includes recipe images, ingredients, preparation instructions, navigation, and a demonstrative contact form. Styling, client-side scripting, server-side processing, databases, authentication, and real message delivery are outside this version.

1.6 User Needs

- Simple navigation between website pages.
- Readable recipe names, descriptions, ingredients, and instructions.

- Clear images with text alternatives.
- A predictable page structure.
- A simple way to view contact information/form fields.

1.7 Constraints and Assumptions

- HTML5 is the primary implementation technology for Version 1.0.
- No CSS or JavaScript is required in this project version.
- No database or backend service is used.
- The contact form is an interface demonstration and does not persist or transmit messages.
- The deployed site depends on a modern browser and internet access.

INTRODUCTION TO PROPOSED SYSTEM

2.1 Introduction

The proposed system is a static website that allows visitors to browse a small collection of recipes and open detailed recipe pages. The website is served as HTML and image files and requires no application server or database.

2.2 Proposed Solution

The solution organizes recipe content into separate, linked HTML pages. A common navigation structure connects the major sections. Recipe detail pages present ingredients as lists and preparation steps in sequence, providing a clear and educational example of semantic web content.

2.3 Target Users

User	Description
Visitor	Browses recipes and website information.
Student Developer	Studies HTML structure, requirements, testing, and deployment.
Reviewer / Instructor	Evaluates whether implementation satisfies documented requirements.

2.4 Product Features

Feature ID	Feature Name
FID01	Home Page
FID02	Recipe Catalog
FID03	Recipe Detail Pages
FID04	Ingredients and Instructions
FID05	About Page
FID06	Contact Form Interface
FID07	Global Navigation
FID08	Accessible Images and Labels

2.5 Operating Environment

- Development environment: Windows 11 and Visual Studio Code.
- Version control: Git with a GitHub repository.

- Client environment: current desktop or mobile web browser.
- Production environment: GitHub Pages or equivalent static hosting over HTTPS.

2.6 System Context

The visitor interacts with the website through a web browser. The hosting platform serves static HTML and image assets. No runtime database, API, or authentication service is required.

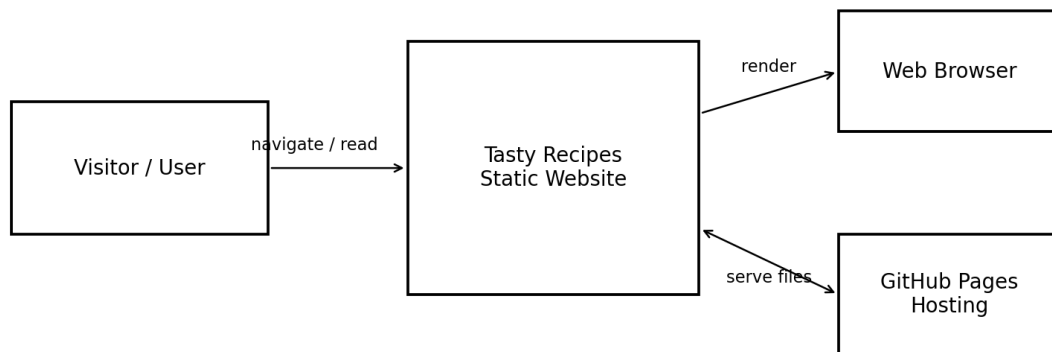


Figure 2.1: System Context Diagram

REQUIREMENT SPECIFICATIONS

3.1 Introduction

This chapter defines the functional, interface, and quality requirements that the implemented website must satisfy. Mandatory requirements use the word SHALL so that each statement can be verified during testing.

3.2 Functional Requirements

Requirement ID	Requirement Statement
FR_01	The system SHALL provide index.html as the website entry page.
FR_02	The system SHALL provide navigation to Home, Recipes, About, and Contact on each top-level page.
FR_03	The system SHALL display at least three recipe summaries on the Recipes page.
FR_04	The system SHALL provide a separate details page for each listed recipe.
FR_05	Each recipe details page SHALL display the recipe name and at least one relevant image.
FR_06	Each recipe details page SHALL display ingredients as a structured list.
FR_07	Each recipe details page SHALL display preparation instructions in ordered sequence.
FR_08	The system SHALL provide an About page describing the website purpose.
FR_09	The Contact page SHALL provide name, email, and message fields with visible labels.
FR_10	Content images SHALL include meaningful alternative text.
FR_11	All local navigation links and required asset paths SHALL resolve correctly.
FR_12	The website SHALL include a footer on all top-level pages.

3.3 Fully Dressed Use Cases

3.3.1 UC-01: Browse Recipes

Item	Description
UC-ID	UC-01

Use Case Name	Browse Recipes
Primary Actor	Visitor
Brief Description	Visitor opens the recipe catalog to view available recipes.
Preconditions	Website is accessible.
Trigger	Visitor selects Recipes from navigation.
Basic Flow	1. System opens recipes.html. 2. System displays recipe summaries. 3. Visitor reviews available recipes.
Postcondition	Recipe catalog remains available for selection.

3.3.2 UC-02: View Recipe Details

Item	Description
UC-ID	UC-02
Use Case Name	View Recipe Details
Primary Actor	Visitor
Brief Description	Visitor opens one recipe to read complete information.
Preconditions	Recipe catalog is available.
Trigger	Visitor selects a recipe link.
Basic Flow	1. System opens selected recipe page. 2. System displays title and image. 3. System displays ingredients. 4. System displays ordered instructions.
Postcondition	Visitor can read the complete recipe and return to the catalog.

3.3.3 UC-03: View About Information

Item	Description
UC-ID	UC-03
Use Case Name	View About Information
Primary Actor	Visitor
Brief Description	Visitor reads the purpose of the website.
Preconditions	Website is accessible.
Trigger	Visitor selects About.
Basic Flow	System opens about.html and displays project/site information.
Postcondition	About information is visible.

3.3.4 UC-04: Use Contact Form

Item	Description
UC-ID	UC-04
Use Case Name	Use Contact Form Interface

Primary Actor	Visitor
Brief Description	Visitor enters information into the contact form fields.
Preconditions	Contact page is open.
Trigger	Visitor selects a form field.
Basic Flow	1. Visitor enters name. 2. Visitor enters email. 3. Visitor enters message. 4. Visitor can activate the submit control.
Alternative Flow	Browser validation may request completion of required fields.
Postcondition	Input is not stored or transmitted by Version 1.0.

3.4 External Interface Requirements

ID	Requirement
IR_01	Every top-level page SHALL display a clear page or site heading.
IR_02	Navigation text SHALL clearly identify its destination.
IR_03	Form fields SHALL have associated text labels.
IR_04	Recipe content SHALL distinguish ingredients from instructions.
IR_05	The deployed website SHALL be accessible through HTTPS.

3.5 Non-Functional Requirements

NFR ID	Category	Description
NFR01	Usability	A visitor SHOULD reach any top-level page using the main navigation.
NFR02	Accessibility	Semantic landmarks, logical headings, labels, and meaningful alt text SHALL be used where applicable.
NFR03	Compatibility	The site SHALL render without critical content loss in current Chrome and Edge.
NFR04	Maintainability	Files SHALL use meaningful names and an organized directory structure.
NFR05	Reliability	The release SHALL contain no known broken internal links or required local images.
NFR06	Security	Passwords, tokens, private keys, and secrets SHALL NOT be committed to the repository.

NFR07	Privacy	Version 1.0 SHALL NOT persist visitor contact information.
NFR08	Portability	The website SHALL deploy on a standard static hosting service.

3.6 Acceptance Criteria

- All mandatory functional requirements are implemented.
- All top-level navigation links work.
- All recipe detail links open the intended page.
- Required images load and include alternative text.
- Contact fields and labels are present.
- No secrets are committed to the repository.
- The production site opens successfully over HTTPS.

DESIGN SPECIFICATIONS

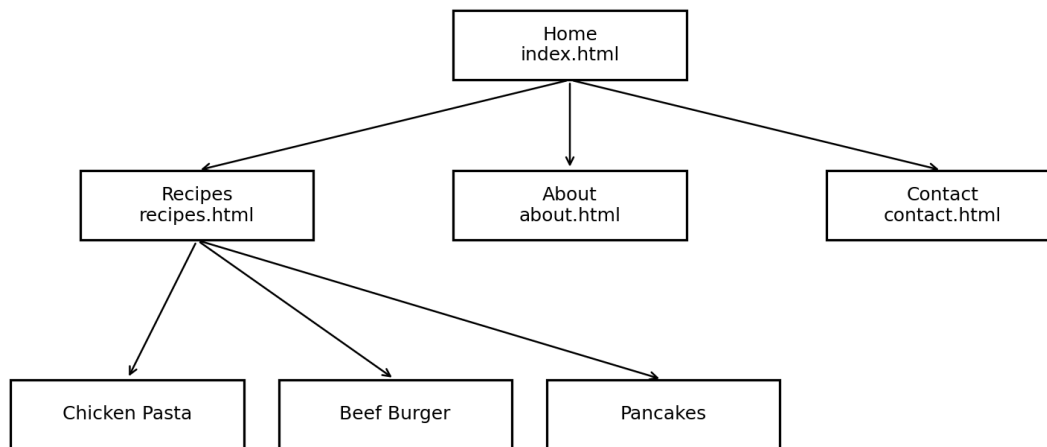
4.1 Introduction

This chapter translates the requirements into a simple static-web design. Because Version 1.0 contains no backend or database, the design focuses on navigation, page hierarchy, content organization, and file structure.

4.2 System Architecture

The architecture consists of three elements: the visitor, a web browser, and static files served by the hosting platform. The browser requests HTML documents and image assets; the hosting platform returns those files. There is no server-side application logic.

4.3 Website Structure



Primary navigation is available from all top-level pages.

Figure 4.1: Website Navigation / Site Map

4.4 Use Case Diagram

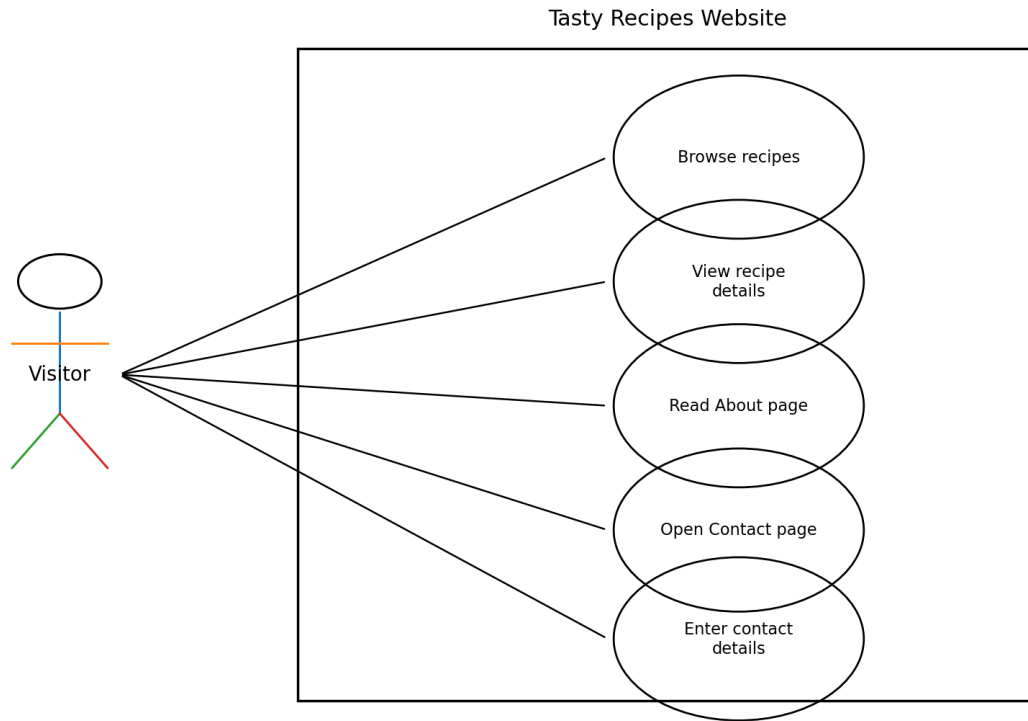


Figure 4.2: Use Case Diagram

4.5 Block Diagram

Tasty Recipes - High-Level Block Diagram

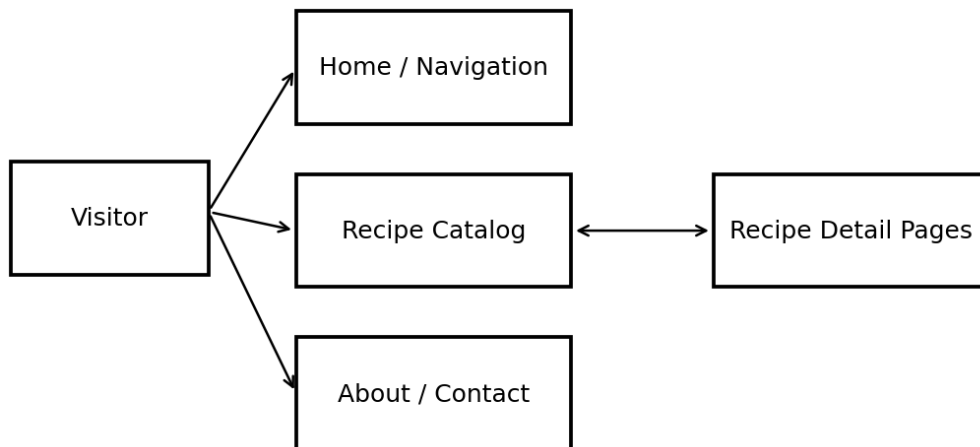


Figure 4.3: High-Level Block Diagram

4.6 Class Diagram

Conceptual Class Diagram

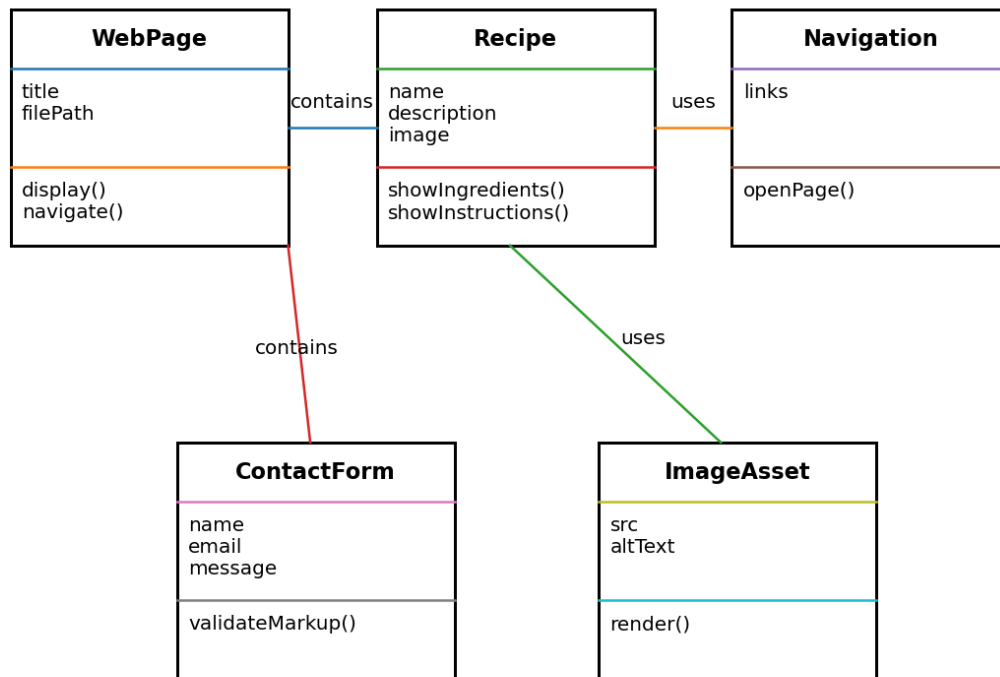


Figure 4.4: Conceptual Class Diagram

4.7 Sequence Diagram

Sequence Diagram - View Recipe Details

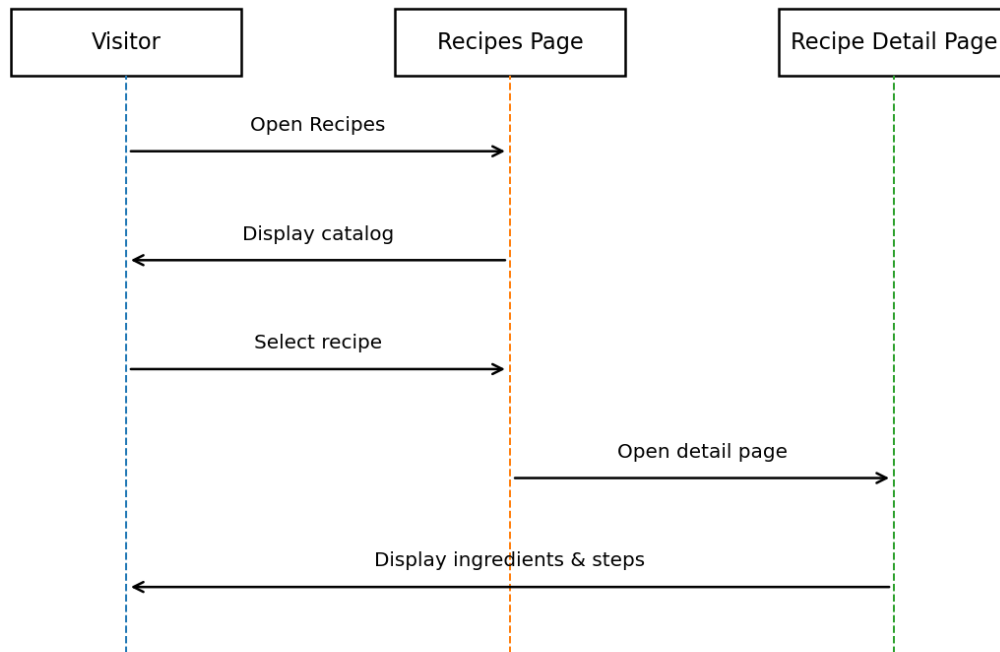


Figure 4.5: Sequence Diagram - View Recipe Details

4.8 File and Folder Design

```
01-tasty-recipes-html/  
├── index.html  
├── recipes.html  
├── about.html  
├── contact.html  
├── recipes/  
│   ├── chicken-pasta.html  
│   ├── beef-burger.html  
│   └── pancakes.html  
├── assets/  
│   └── images/  
├── docs/  
│   └── Tasty_Recipes_SRS.docx  
└── README.md
```

TEST SPECIFICATIONS

5.1 Introduction

Testing verifies that the implemented HTML satisfies the documented requirements and that the deployed website behaves consistently for its intended static functionality.

5.2 Test Strategy

- Functional testing: verify pages, links, images, lists, and form controls.
- Compatibility testing: open the website in current Chrome and Edge.
- Accessibility review: verify heading order, labels, semantic structure, and image alt text.
- Deployment testing: repeat critical navigation checks on the live hosted URL.

5.3 Test Cases

Test ID	Test Description	Test Action	Expected Result	Requirement
TC_01	Home Page	Open the deployed root URL.	Home page loads and displays website identity.	FR_01
TC_02	Navigation	Use all top-level navigation links.	Each link opens the correct page.	FR_02, FR_11
TC_03	Recipe Catalog	Open Recipes.	At least three recipe summaries are displayed.	FR_03
TC_04	Recipe Details	Open each listed recipe.	Correct page displays image, ingredients, and instructions.	FR_04-FR_07
TC_05	Contact Form	Open Contact and inspect controls.	Name, email, message, labels, and submit control are present.	FR_09
TC_06	Alternative Text	Inspect content images.	Meaningful alt text exists.	FR_10
TC_07	Browser Compatibility	Open site in Chrome and Edge.	No critical content/navigation failure occurs.	NFR03
TC_08	Repository Security	Review tracked files before release.	No passwords, tokens, or private keys are committed.	NFR06
TC_09	Deployment	Open live site using HTTPS.	Production website is reachable and navigation works.	NFR08, IR_05

5.4 Requirements Traceability Matrix

Requirement	Design / Implementation	Verification
FR_01	index.html	TC_01
FR_02	Navigation on top-level pages	TC_02
FR_03	recipes.html	TC_03
FR_04-FR_07	recipes/*.html	TC_04
FR_09	contact.html	TC_05
FR_10	Image alt attributes	TC_06
NFR03	All pages	TC_07
NFR06	Repository	TC_08
NFR08 / IR_05	GitHub Pages deployment	TC_09

DEPLOYMENT AND CONCLUSION

6.1 Deployment Requirements

- Source code shall be maintained in the GitHub repository.
- The production release shall be based on the main branch.
- The deployed entry point shall resolve to index.html.
- All relative links and asset paths shall work from the production base URL.
- The live deployment link shall be added to README.md after release.

6.2 Deployment Procedure

- Complete implementation and local testing.
- Review git status and stage only intended files.
- Commit the release-ready version with a meaningful commit message.
- Push the main branch to GitHub.
- Enable GitHub Pages for the repository.
- Open the live URL and repeat critical test cases.
- Add the verified live URL to README.md.

6.3 Limitations

- No visual styling beyond browser defaults in Version 1.0.
- No JavaScript interactions.
- No backend or database.
- Contact form does not send or store messages.
- Recipe content is manually maintained in HTML files.

6.4 Future Enhancements

- Responsive CSS design.
- JavaScript search and filtering.
- Client-side form validation and interactive features.
- Backend contact processing.
- MySQL-based recipe storage.
- Authentication and user accounts.
- Admin recipe management and REST API integration.

6.5 Conclusion

Tasty Recipes provides a controlled first project for learning the complete workflow from requirements to design, implementation, testing, version control, and deployment. The limited HTML-only scope keeps

the technical focus clear while establishing documentation and engineering practices that can be reused in larger web, backend, database, and desktop projects.

Chapter A

GLOSSARY

Term	Definition
HTML5	Markup language used to structure web content.
Static Website	A website served as prebuilt files without server-side application logic.
Semantic HTML	HTML elements selected according to the meaning of the content.
Functional Requirement	A behavior or capability the system must provide.
Non-Functional Requirement	A quality attribute or constraint applied to the system.
Use Case	A structured description of an actor achieving a goal with the system.
Acceptance Criteria	Conditions used to determine whether the implementation is acceptable.
Git	Distributed version-control system.
GitHub Pages	Static hosting service used to publish repository content.

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- [1] ISO/IEC/IEEE 29148:2018, Systems and software engineering - Life cycle processes - Requirements engineering.
- [2] WHATWG, HTML Living Standard.
- [3] W3C, Web Content Accessibility Guidelines (WCAG).